Exploratory Data Analysis (EDA) Report

E-Commerce Sales & Customer Behaviour Analysis

1. Project Objectives

This project performs an in-depth Exploratory Data Analysis (EDA) on an e-commerce sales dataset to understand its structure, assess data quality, and uncover meaningful business patterns. The goal is to support data-driven decision-making through insights into sales, profitability, customer behavior, product performance, and regional trends.

- Investigate the structure and quality of the e-commerce dataset.
- Identify missing values, duplicate records, and inconsistencies affecting reliability.
- Clean and pre-process the data to improve its analytical quality.
- Analyse the distribution of key variables: Sales, Profit, Quantity, Discount, and Shipping Cost.
- Evaluate customer, product, and regional performance.
- Quantify relationships between business variables through correlation analysis.
- Translate findings into actionable business recommendations.

2. Business Problem

E-commerce organizations generate large volumes of transactional data daily. Raw business data often contains duplicate records, missing values, and inconsistent entries that can lead to inaccurate reporting and poor decisions. This project cleans and structures the available sales dataset, then explores it to identify significant business patterns — helping decision-makers understand customer purchasing behaviour, product performance, regional trends, profitability drivers, and operational efficiency.

3. Dataset Description

The dataset contains transactional e-commerce records. The original file has 100,500 records and 16 variables. After removing 500 duplicate Order_ID entries, the cleaned dataset contains 100,000 unique transactions covering customer details, product information, order dates, regions, sales metrics, discounts, shipping costs, and payment methods.

Dataset Variables

Column	Type	Business Meaning
Order_ID	Integer	Unique identifier for each order.
Order_Date	Object → Datetime	Date the order was placed.
Ship_Date	Object → Datetime	Date the order was shipped.
Customer_ID	Integer	Unique identifier for each customer.
Customer_Segment	Object	Consumer, Corporate, or Home Office.
Region	Object	Geographic region: East, West, South, North.
State	Object	State associated with the order.
Product_Category	Object	Electronics, Furniture, or Clothing.

Sub_Category	Object	Specific product classification.
Product_Name	Object	Name of the purchased product.
Sales	Float	Revenue generated from the transaction.
Quantity	Integer	Number of units purchased.
Discount	Float	Discount rate applied to the order.
Profit	Float	Profit earned after costs.
Shipping_Cost	Float	Cost incurred to ship the order.
Payment_Mode	Object	Payment method used by the customer.

Data quality snapshot: Customer_Segment had 3,019 missing values, Profit had 3,019 missing values, and Shipping_Cost had 3,006 missing values prior to cleaning. All other fields were fully populated.

4. Data Cleaning & Preprocessing

Data cleaning improves the quality, consistency, and reliability of the dataset before analysis. The following steps were applied in sequence.

4.1 Removing Duplicate Records

Purpose: Eliminate repeated transactions using the unique Order_ID, so every order is represented once.

```
df.duplicated(subset="Order_ID").sum() df.drop_duplicates(subset="Order_ID", inplace=True)
```

Explanation: 500 duplicate Order_ID records were found and removed, keeping the first occurrence of each order and leaving 100,000 unique transactions.

Business Impact: Prevents inflated sales, profit, and transaction counts — leading to accurate business reporting.

4.2 Handling Missing Values

Purpose: Replace missing values with statistically appropriate estimates while preserving the overall distribution.

```
df["Customer_Segment"] = df["Customer_Segment"].fillna(df["Customer_Segment"].mode()[0])
df["Shipping_Cost"] = df["Shipping_Cost"].fillna(df["Shipping_Cost"].median()) df["Profit"] =
df["Profit"].fillna(df.groupby("Sub_Category")["Profit"].transform("mean"))
```

Explanation: Customer_Segment (3,019 missing) was filled with the mode; Shipping_Cost (3,006 missing) with the median; Profit (3,019 missing, after outlier treatment) with the mean profit of the matching Sub_Category.

Business Impact: Preserves valuable transaction records and reduces bias compared to dropping rows.

4.3 Data Type Conversion

Purpose: Convert date columns into proper datetime format for time-based analysis.

```
df["Order_Date"] = pd.to_datetime(df["Order_Date"]) df["Ship_Date"] =
pd.to_datetime(df["Ship_Date"])
```

Explanation: Order_Date and Ship_Date were converted from text (object) to datetime, enabling extraction of month, year, quarter, and shipping duration.

Business Impact: Enables trend, seasonal, and shipping-performance analysis.

4.4 Logical Validation

Purpose: Identify and correct logically invalid values that could distort results.

```
df["Shipping_Cost"] = df["Shipping_Cost"].replace(10000, 100)
```

Explanation: An abnormal Shipping_Cost value (10,000) was identified as a data-entry error and replaced with a realistic value.

Business Impact: Prevents a single erroneous entry from distorting summary statistics.

4.5 Standardization

Purpose: Maintain consistent formatting of categorical text values.

```
df["Product_Category"] = df["Product_Category"].str.strip().str.title()
```

Explanation: Product_Category contained inconsistent casing (e.g., "ELECTRONICS", "electronics", "Electronics"). Extra spaces were stripped and text converted to title case.

Business Impact: Prevents duplicate category labels and improves grouping, filtering, and visualization accuracy.

4.6 Feature Engineering

Purpose: Derive additional variables that support deeper exploratory analysis.

```
df["Month"] = df["Order_Date"].dt.strftime("%b") df["Year"] = df["Order_Date"].dt.year
df["Month_num"] = df["Order_Date"].dt.month df["Shipping_days"] = df["Ship_Date"] -
df["Order_Date"] df["Quarter"] = pd.cut(df["Month_num"], bins=[0,3,6,9,12],
labels=["Q1","Q2","Q3","Q4"])
```

Explanation: New fields — Month, Year, Month_num, Shipping_days, and Quarter — were derived from the order/ship dates.

Business Impact: Provides the dimensions needed to detect seasonal demand and logistics patterns.

4.7 Outlier Handling (Profit)

Purpose: Detect extreme observations and reduce their influence on statistical analysis.

```
Q1, Q3 = df["Profit"].quantile([0.25, 0.75]) IQR = Q3 - Q1 lower, upper = Q1 - 1.5*IQR, Q3 +
1.5*IQR df["Profit"] = df["Profit"].where(df["Profit"].between(lower, upper))
```

Explanation: The IQR method flagged extreme Profit values, which were converted to missing and later re-imputed using the average profit of the matching Sub_Category (Step 4.2).

Business Impact: Reduces the influence of extreme values while retaining every transaction record.

4.8 Post-Cleaning Validation

Purpose: Confirm that all corrections and imputations were applied successfully.

```
df["Shipping_Cost"].sort_values().unique() df.info()
```

Explanation: Validation checks confirmed the dataset was free of the previously identified issues and ready for analysis.

Business Impact: Ensures the dataset is accurate, consistent, and trustworthy before generating business insights.

4.9 Data Cleaning Summary

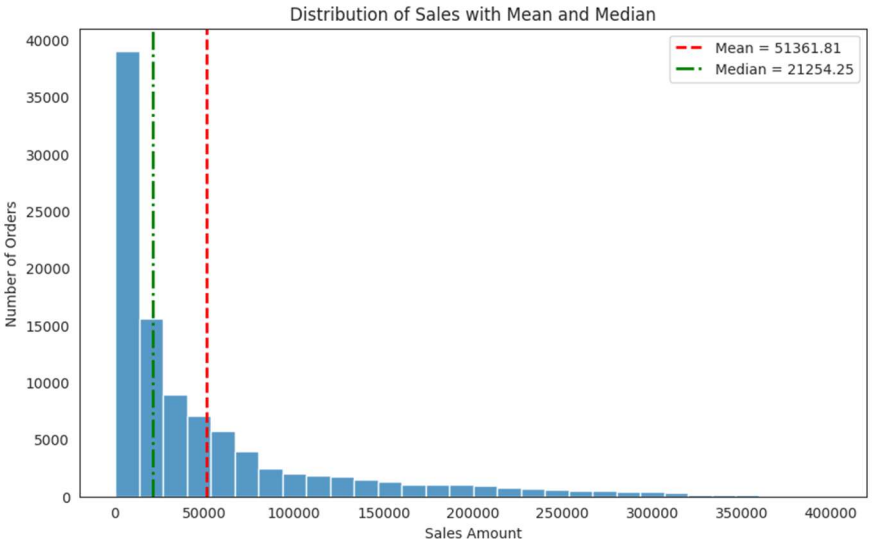
Cleaning Activity	Status	Result
Duplicate Removal	✓ Completed	500 duplicates removed
Missing Value Treatment	✓ Completed	Segment / Profit / Shipping_Cost imputed
Date Type Conversion	✓ Completed	Order_Date, Ship_Date → datetime
Logical Value Validation	✓ Completed	Invalid shipping cost corrected
Text Standardization	✓ Completed	Product_Category normalized
Feature Engineering	✓ Completed	Month, Year, Quarter, Shipping_days added
Outlier Treatment (Profit)	✓ Completed	IQR method + sub-category mean imputation
Final Dataset Size	100,000 rows	16 → 21 columns (incl. engineered features)

Note on outlier methodology: only the IQR method was used for Profit outlier treatment. A Z-score approach was not applied, since Sales, Profit, and Discount are all right-skewed rather than normally distributed — a condition the Z-score method assumes but IQR does not.

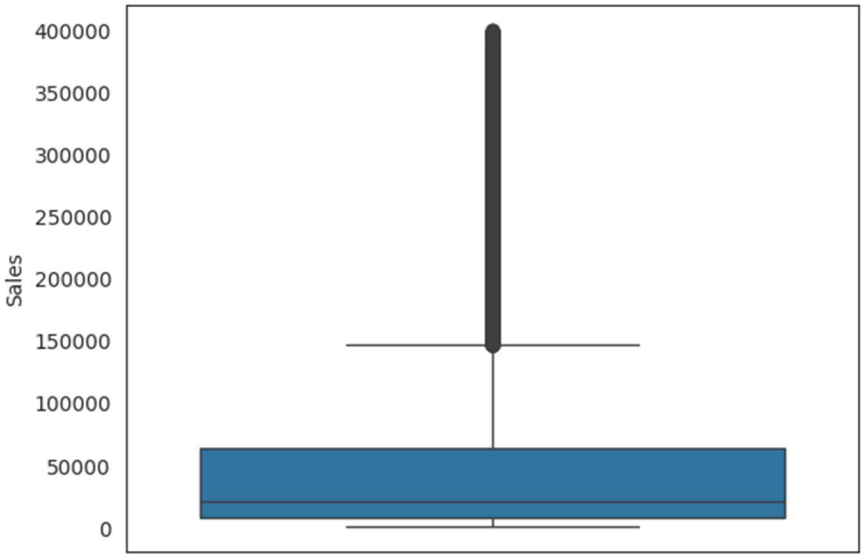
5. Exploratory Data Analysis

5.1 Univariate Analysis

Sales--



Distribution of Sales with mean and median

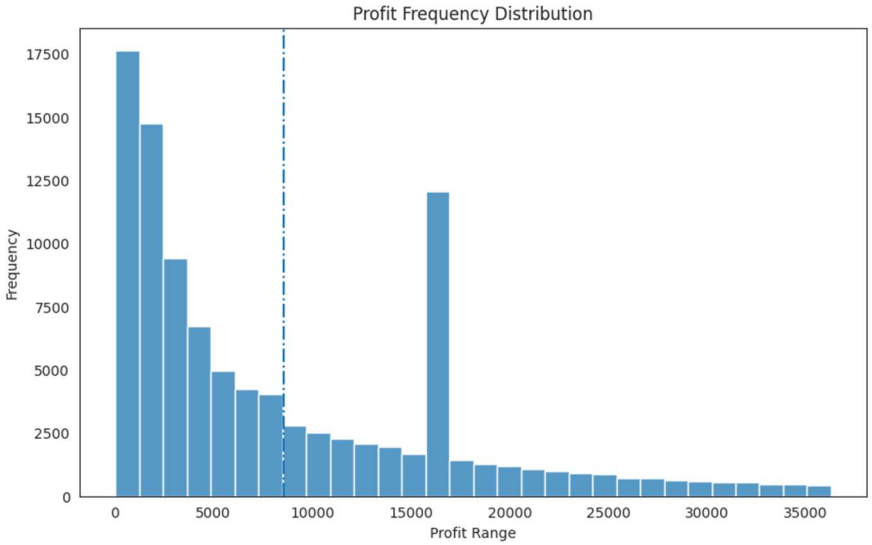


Boxplot of Sales

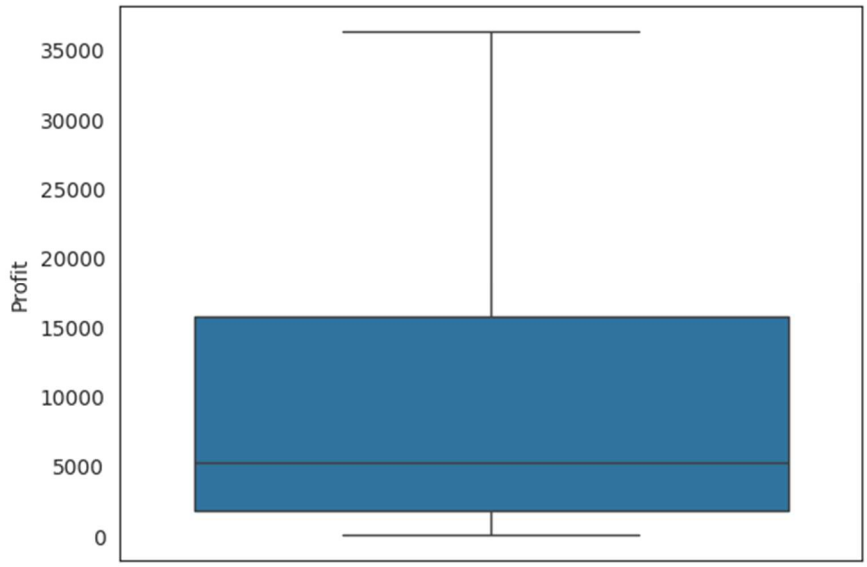
Observation: Sales is highly right-skewed (skewness ≈ 2.13). The mean (₹51,362) sits well above the median (₹21,254), confirming that most orders are low-to-moderate value while a small number of high-value orders pull the average upward.

Business Insight: A small share of large orders drives a disproportionate share of revenue — consistent with typical e-commerce demand patterns.

Profit--



Profit frequency distribution

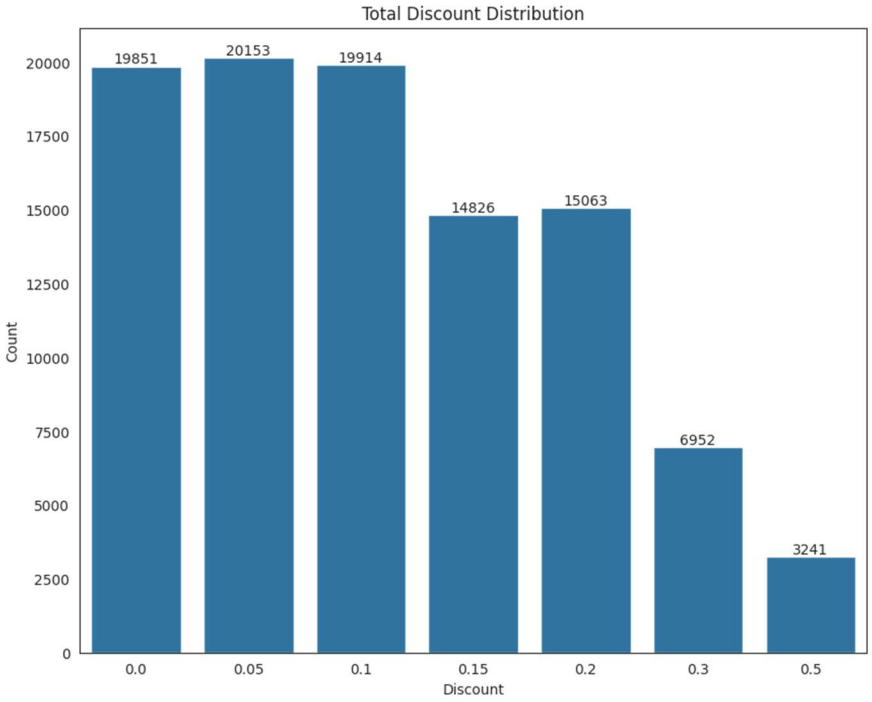


Boxplot of Profit

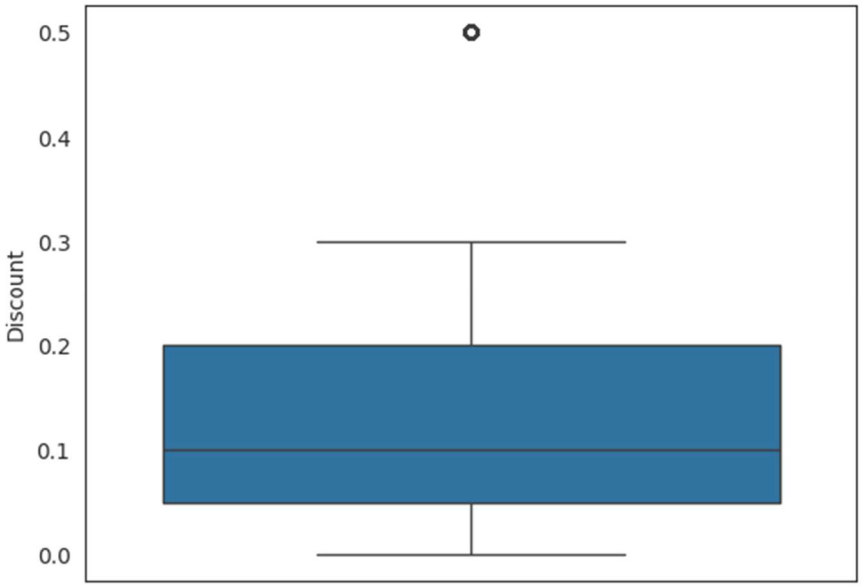
Observation: Profit is also right-skewed, with a large cluster of orders earning modest profit and a secondary peak around ₹16,000–17,000 from repeatedly-imputed sub-category averages. A smaller number of orders generate profits above ₹25,000.

Business Insight: Profitability is uneven across orders; identifying what drives the high-profit segment (product mix, category) can help replicate that performance more broadly.

Discount---



Discount value distribution

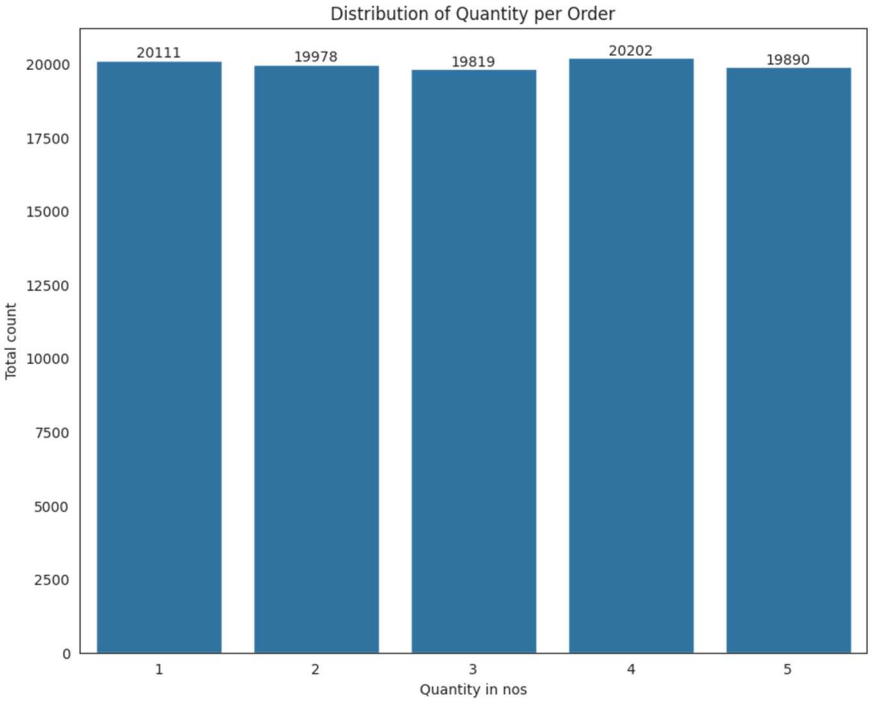


Boxplot of Discount

Observation: Discount is moderately right-skewed (skewness ≈ 1.83). Most orders receive a discount of 10% or less, while discounts of 30% and 50% are applied far less frequently (6,952 and 3,241 orders respectively).

Business Insight: Deep discounts are the exception rather than the rule, which limits their overall drag on profitability (see Section 5.6).

Quantity---

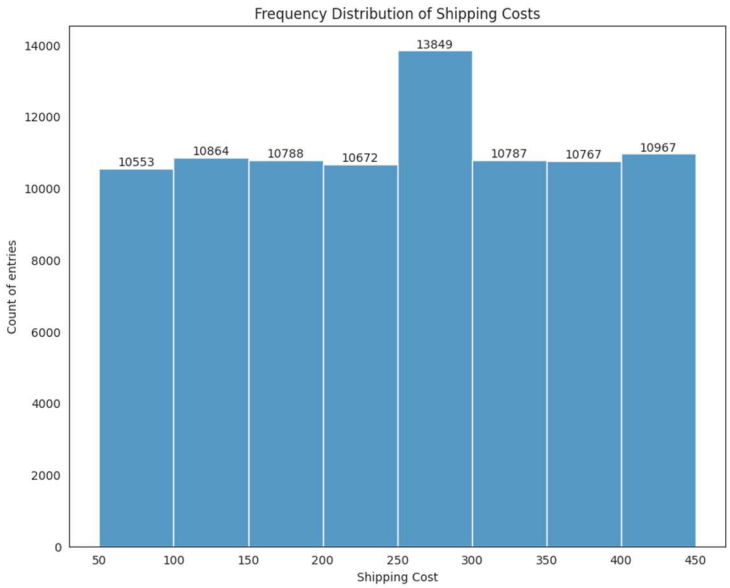


Distribution of quantity per order

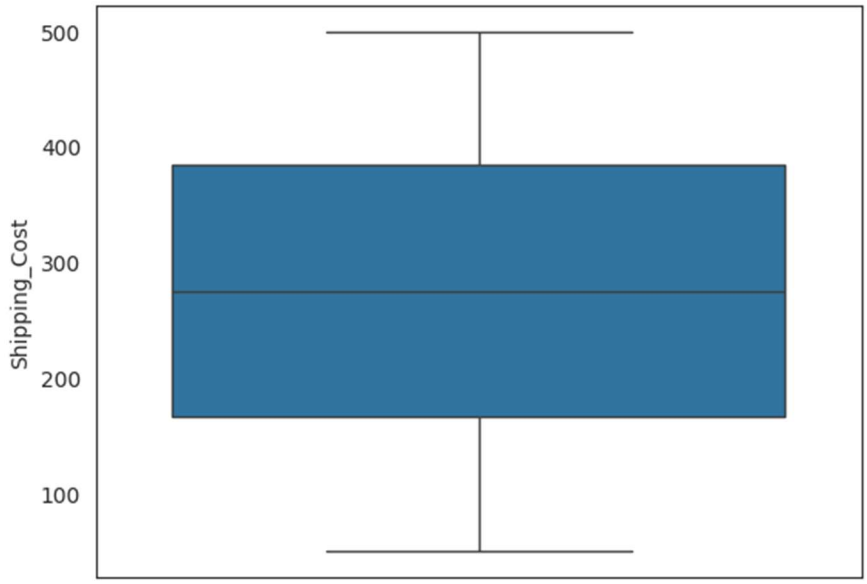
Observation: Quantity is approximately uniform across 1–5 units per order (skewness ≈ 0), with each value accounting for roughly 19,800–20,200 orders.

Business Insight: Purchase quantity is stable and predictable, simplifying inventory and fulfilment planning.

Shipping Cost---



Frequency distribution of shipping cost



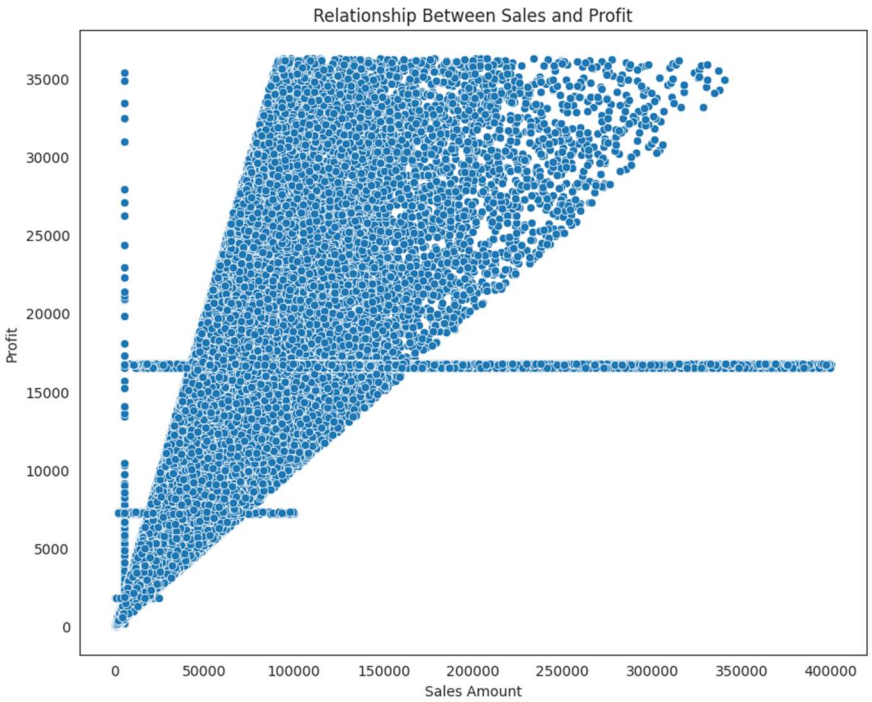
Boxplot of Shipping Cost

Observation: Shipping cost is slightly right-skewed and fairly evenly spread between ₹50 and ₹450, with a concentration of orders in the ₹250–300 band.

Business Insight: Shipping costs are consistent and well-controlled, with no major cost spikes after the Step 4.4 correction.

5.2 Bivariate Analysis

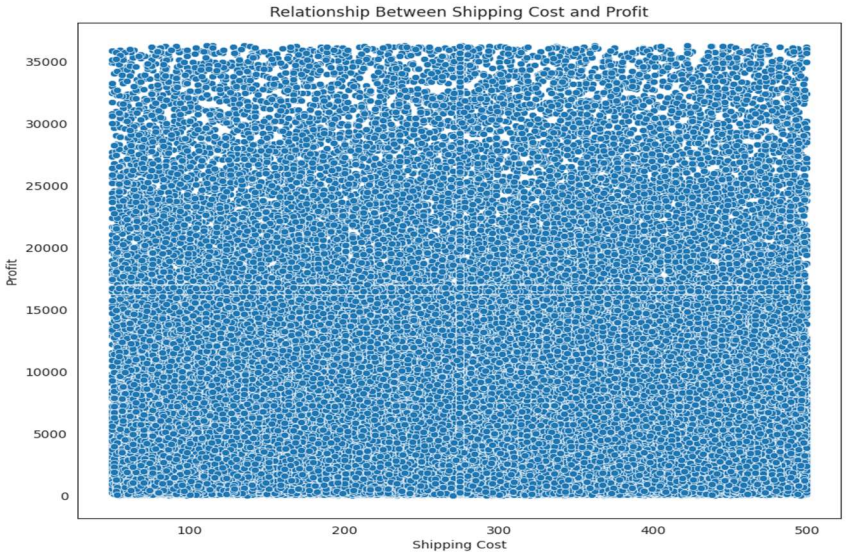
Sales vs. Profit---



Sales vs. Profit

Observation: Profit rises with Sales in a clear, strong, positive pattern — the two variables move together closely, though profit growth plateaus at higher sales values.

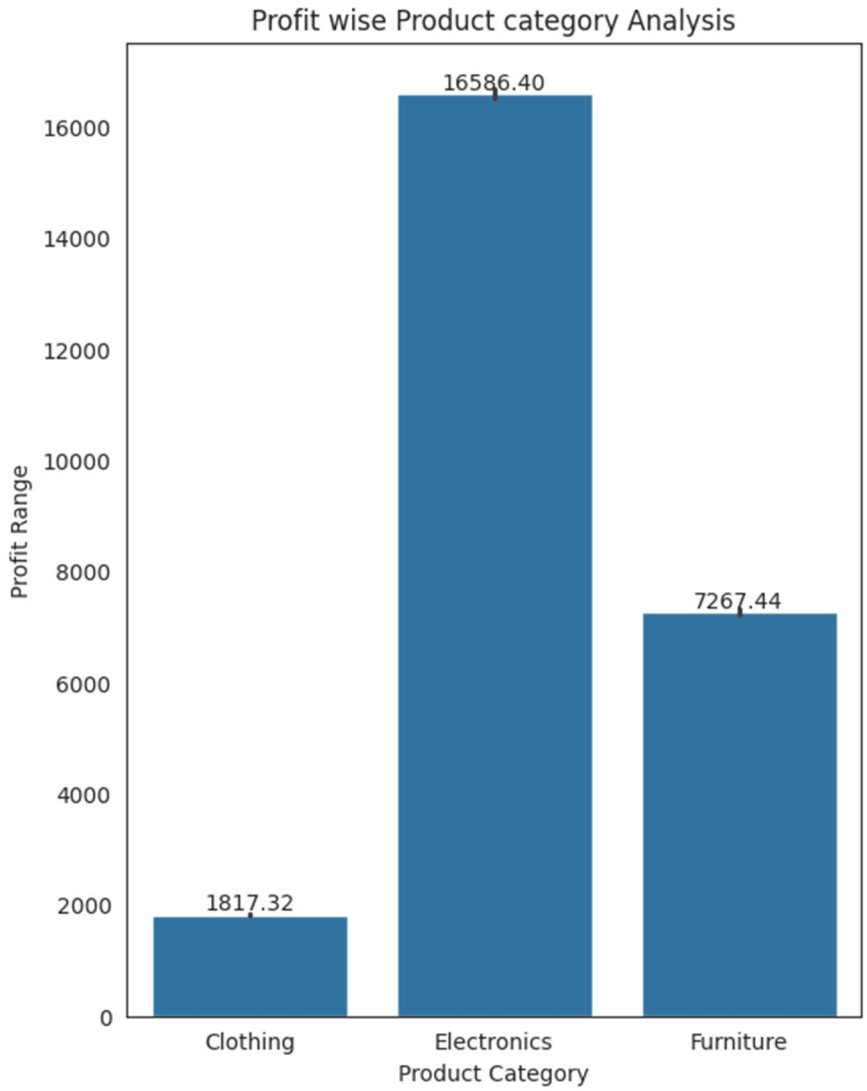
Shipping Cost vs. Profit---



Shipping Cost vs. Profit

Observation: Points are scattered with no visible trend — shipping cost has no meaningful effect on profit.

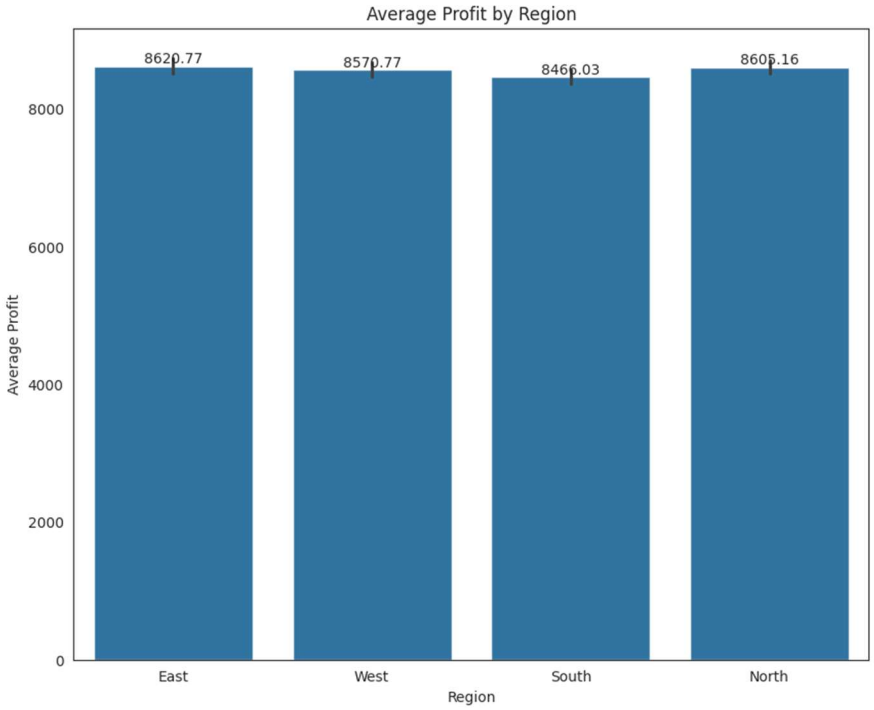
Product Category vs. Profit---



Average profit per order by product category

Observation: Electronics earns by far the highest average profit per order (₹16,586), followed by Furniture (₹7,267) and Clothing (₹1,817).

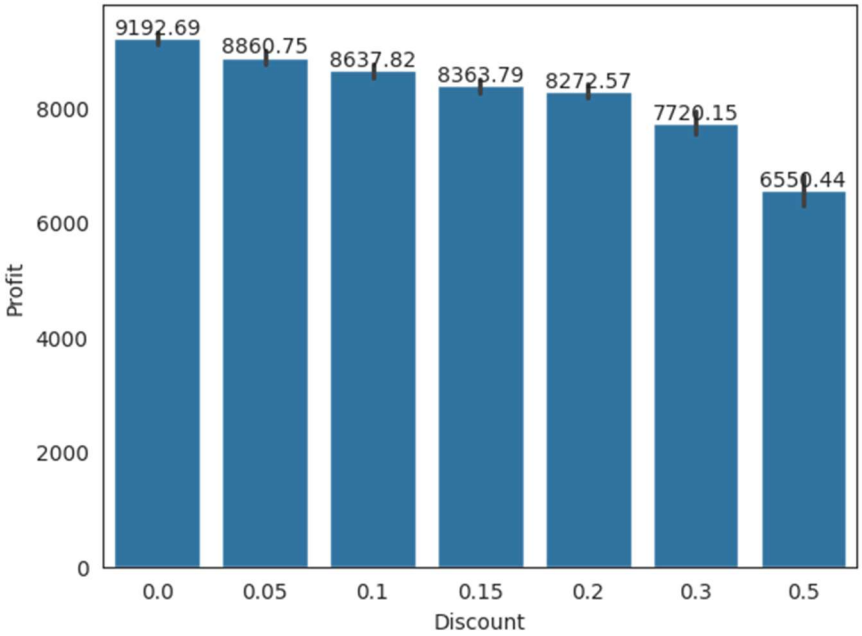
Region vs. Profit---



Average profit per order by region

Observation: Average profit per order is close across regions (East ₹8,621, North ₹8,605, West ₹8,571, South ₹8,466), with the South trailing slightly.

Discount vs. Profit---



Profit by discount level

Observation: Total profit declines steadily as the discount rate increases, confirming that heavier discounting erodes profitability.

Sales – Discount – Profit Correlation---

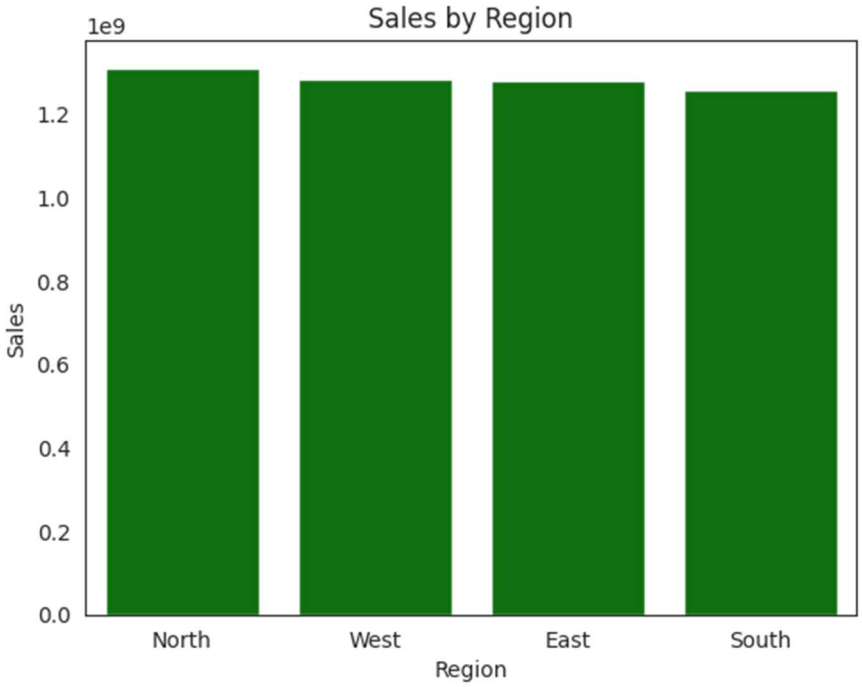


Correlation heatmap: Sales, Discount, Shipping Cost, Profit

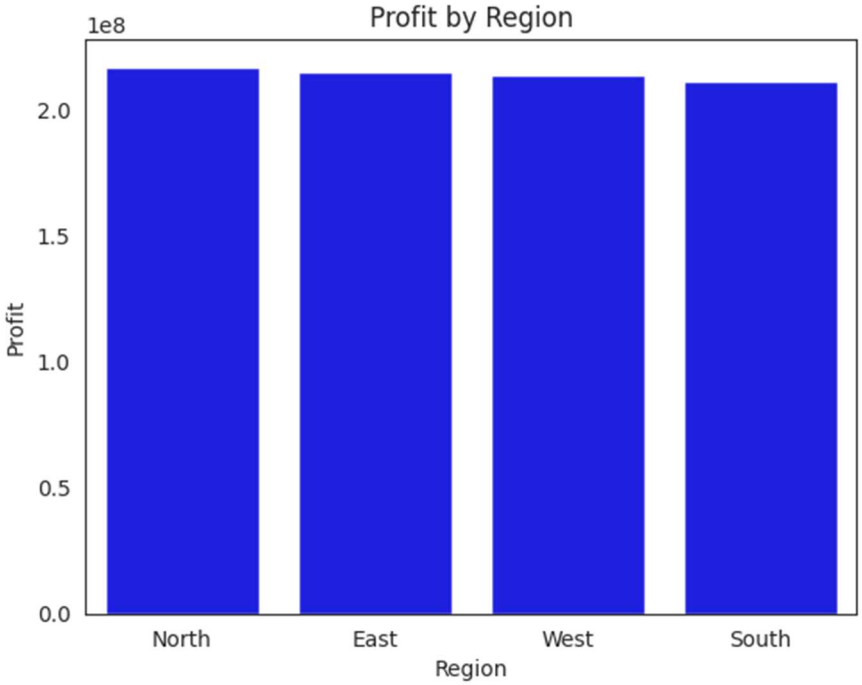
Observation: Sales correlates strongly with Profit (r ≈ 0.71); Discount correlates weakly and negatively with Profit (r ≈ -0.09); Shipping Cost shows virtually no correlation with Profit.

5.3 Categorical Analysis

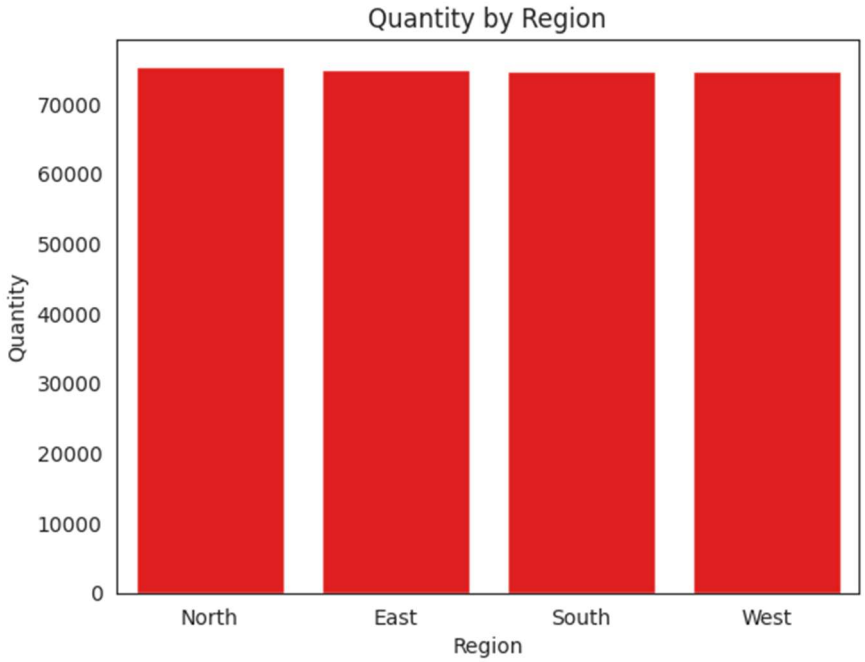
Sales, Profit, Quantity & Shipping Cost by Region---



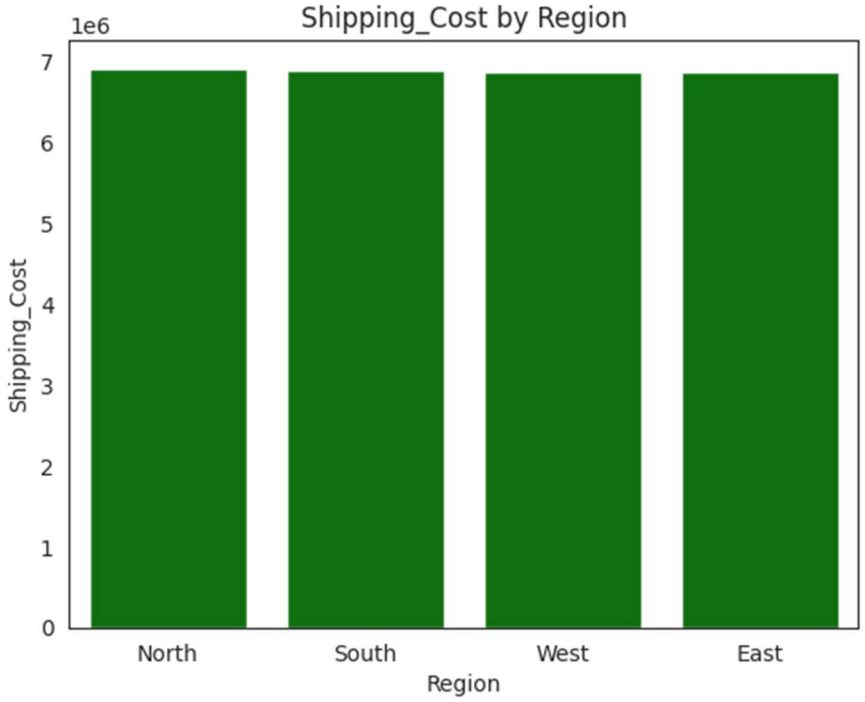
Total sales by region



Total profit by region



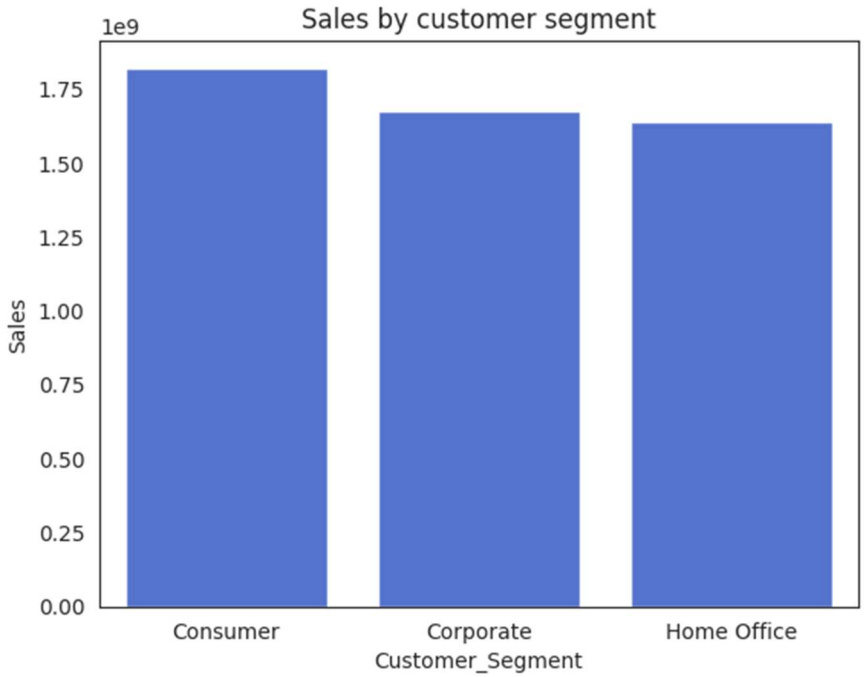
Total quantity sold by region



Total shipping cost by region

Observation: North leads on both total sales (~₹1.31B) and total profit (~₹217M), followed closely by West, East, and South. Quantity sold and total shipping cost are nearly identical across all four regions, indicating balanced order volume nationally.

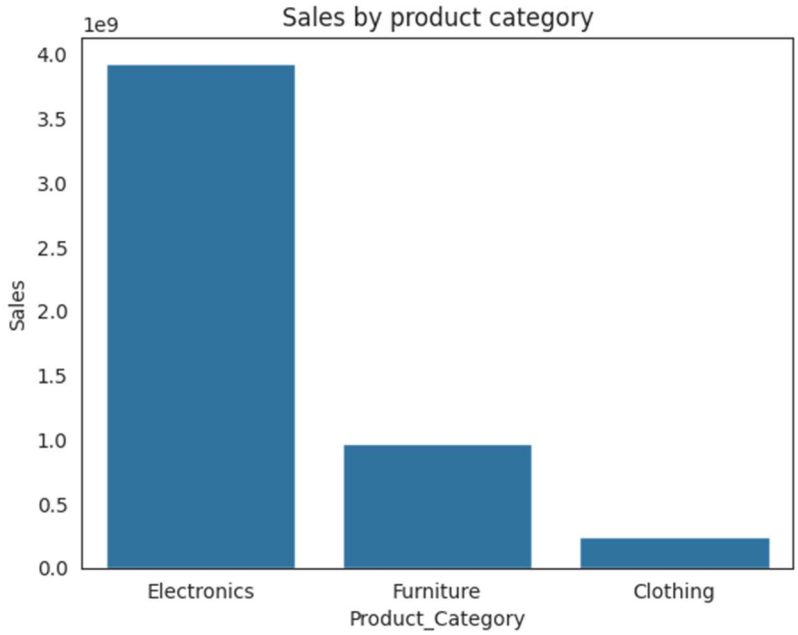
Sales by Customer Segment---



Total sales by customer segment

Observation: Consumer generates the highest total sales (~₹1.82B), ahead of Corporate (~₹1.67B) and Home Office (~₹1.64B).

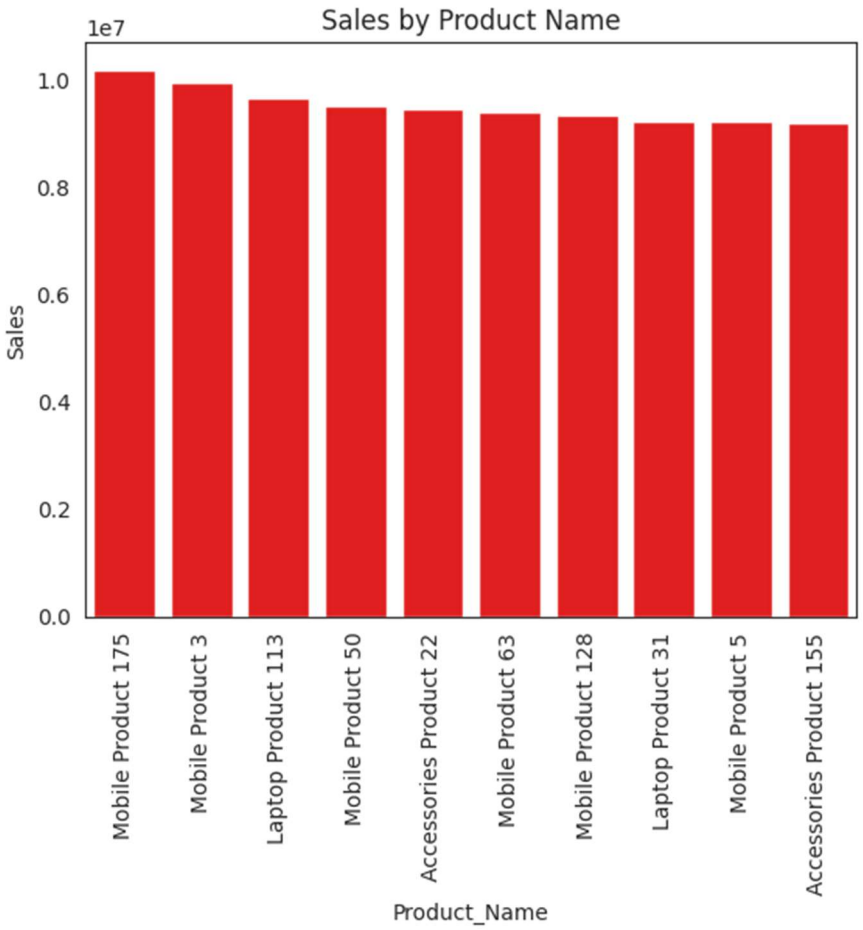
Sales by Product Category---



Total sales by product category

Observation: Electronics dominates revenue (~₹3.9B), roughly four times Furniture (~₹0.97B) and well ahead of Clothing (~₹0.24B).

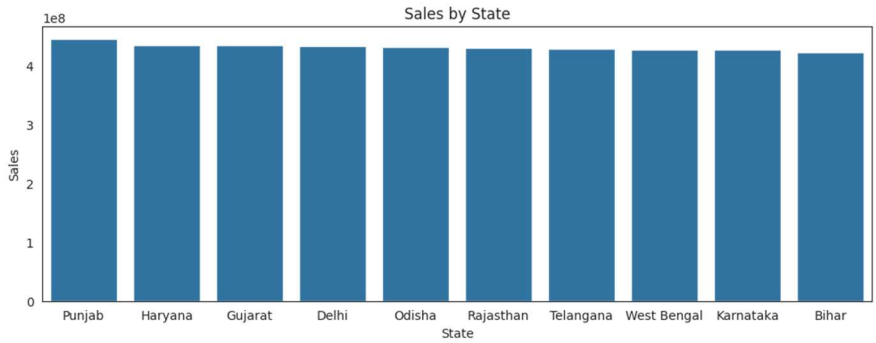
Top 10 Products by Sales---



Top 10 products by total sales

Observation: The top 10 products are led by Mobile Product 175, Mobile Product 3, and Laptop Product 113, with sales fairly evenly spread across the top 10 (no single product dominates).

Top 10 States by Sales---

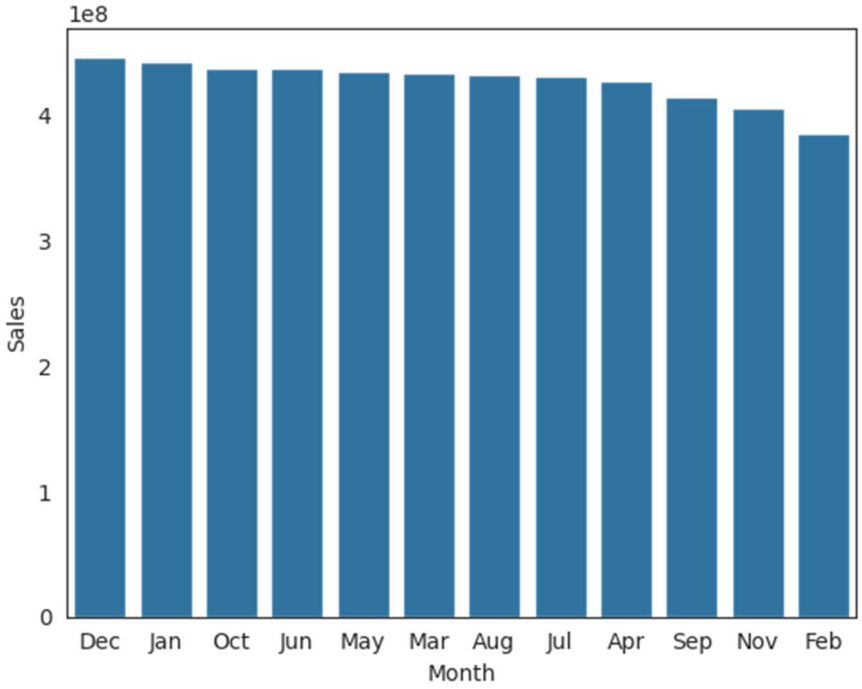


Top 10 states by total sales

Observation: Punjab leads state-level sales, closely followed by Haryana, Gujarat, Delhi, and Odisha — the top 10 states are all within a narrow band, showing broad-based demand rather than concentration in one region.

5.4 Time-Based Analysis

Monthly Sales Trend---



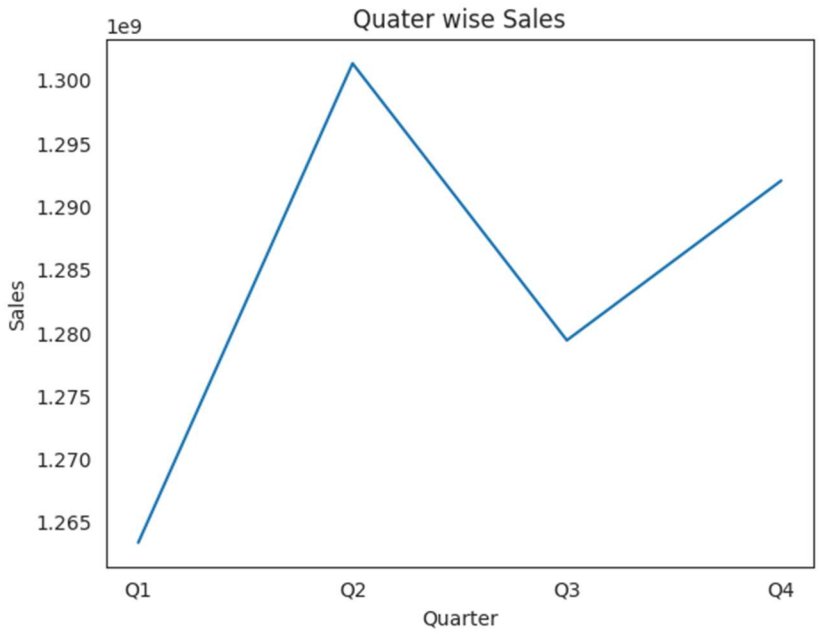
Total sales by month

Observation: December recorded the highest total sales, with January and October also performing strongly. Sales stayed relatively stable from March through August, while February recorded the lowest total sales.

Business Insight: December's peak reflects holiday and year-end purchasing; the post-holiday dip in February suggests reduced demand once festive spending ends.

Recommendation: Increase inventory, staffing, and marketing ahead of December; run targeted promotions in February to lift the seasonal low.

Quarterly Sales Trend---

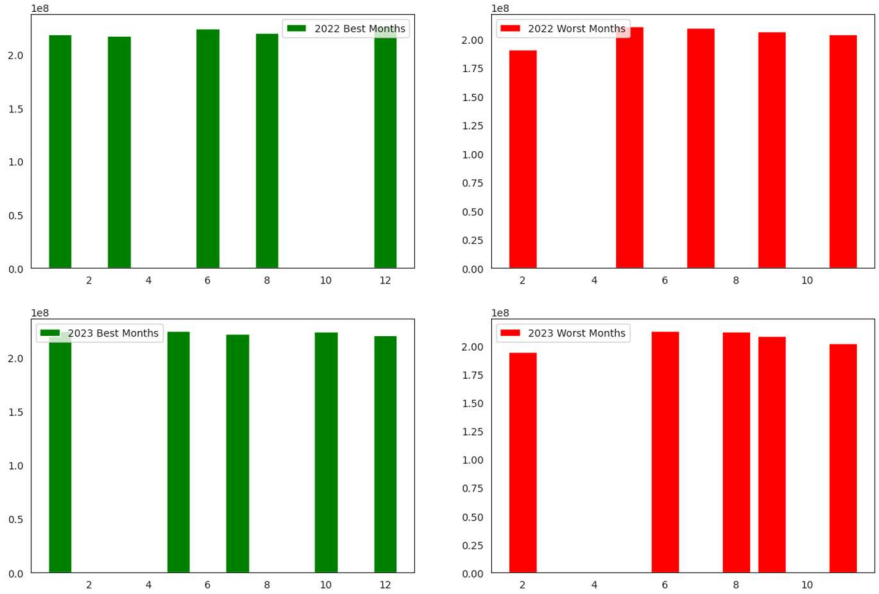


Total sales by quarter

Observation: Q2 recorded the highest sales (~₹1.30B), while Q1 was the weakest (~₹1.26B). Sales dipped in Q3 before recovering in Q4, though Q4 stayed slightly below the Q2 peak.

Recommendation: Replicate the campaigns and product mix that drove Q2's performance in other quarters, and prioritize demand-generation activity in Q1.

Seasonal Patterns & Best/Worst Months (Year-wise)---



Monthly sales by year — 2022 vs. 2023

Observation: In 2022, December, June, August, January, and March were the strongest months, while February, May, July, September, and November were the weakest. In 2023, January, May, July, October, and December were strongest, while February, June, August, September, and November lagged. February was consistently among the weakest months in both years, and December consistently among the strongest — a recurring seasonal pattern.

Recommendation: Use the recurring December peak and February dip to guide inventory planning, marketing calendars, and resource allocation year over year.

5.5 Customer Analysis

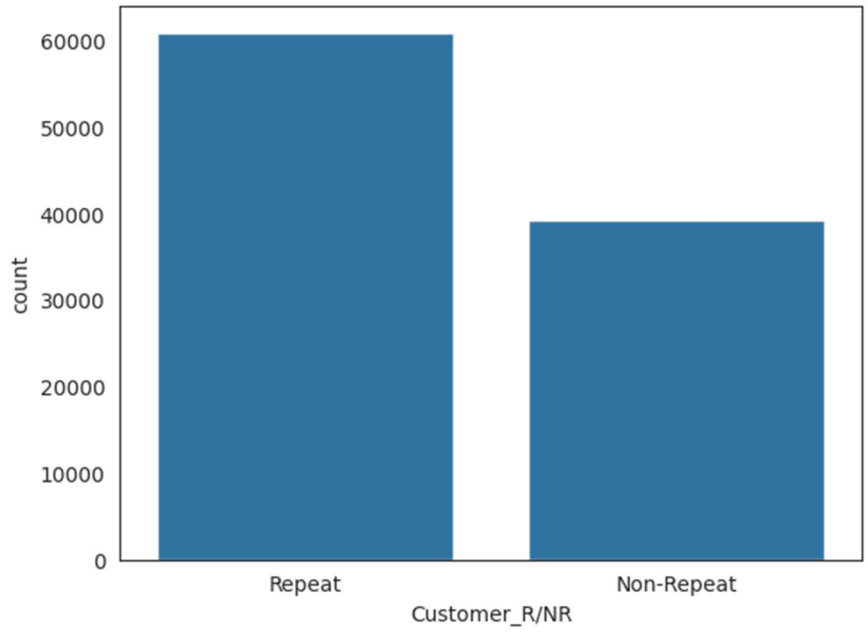
Top 20% Customer Contribution (Pareto Check)

```
t = df.groupby("Customer_ID")["Sales"].sum().sort_values(ascending=False) top20_share =
t.head(int(len(t)*0.2)).sum() * 100 / t.sum()
```

Result: The top 20% of customers by spend contribute approximately 29.8% of total sales.

Business Insight: This falls well short of a classic 80/20 Pareto pattern — revenue is broadly distributed across the customer base rather than concentrated in a small VIP segment, meaning retention strategies should focus on the whole customer base rather than a narrow top tier.

Repeat Customer Behavior---



Repeat vs. non-repeat customers (orders > 24 = repeat)

Observation: The majority of customers are classified as repeat buyers under this threshold, clearly outnumbering non-repeat customers.

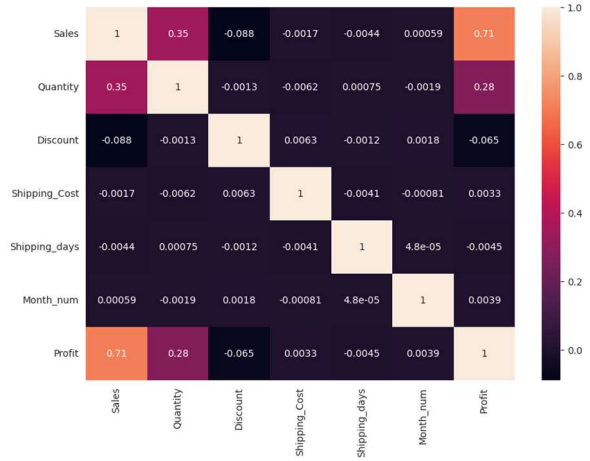
Business Insight: A large repeat-purchase base signals strong customer engagement and an opportunity to grow lifetime value through loyalty programs.

Average Order Value (AOV) by Customer Segment---

Segment	Total Sales	Orders	AOV
Consumer	₹1.82 Billion	35,483	₹51,332
Corporate	₹1.67 Billion	32,136	₹52,077
Home Office	₹1.64 Billion	32,381	₹50,685

Observation: Corporate customers place fewer orders but spend slightly more per order (₹52,077 AOV) than Consumer or Home Office — a smaller, higher-value buying pattern typical of business purchasing.

5.6 Correlation Analysis---



Full correlation matrix: Sales, Quantity, Discount, Shipping Cost, Shipping Days, Month, Profit



Focused view: Sales, Discount, Profit

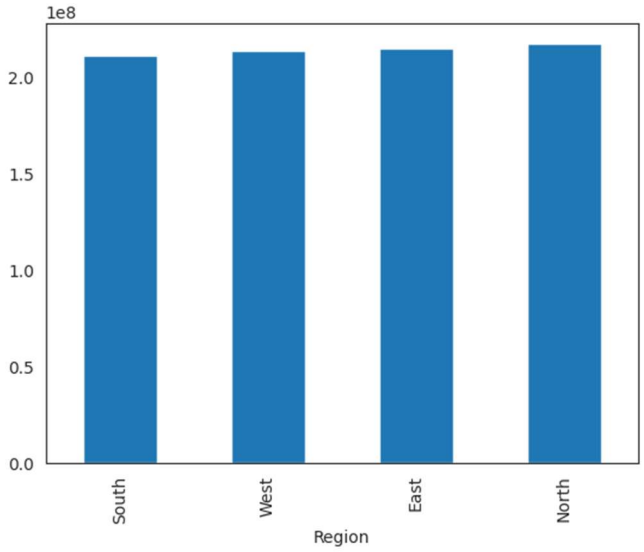
Strength of Relationship with Profit---

Variable	Correlation with Profit	Relationship
Sales	0.709	Strong positive
Quantity	0.276	Weak positive
Discount	-0.065	Very weak negative
Shipping_Cost	0.003	No relationship
Shipping_days	-0.004	No relationship
Month_num	0.004	No relationship

Observation: Sales is by far the strongest driver of Profit. Quantity has a modest positive influence. Discount, Shipping_Cost, shipping duration, and month of order show no meaningful relationship with profit — a Z-score approach was not used here for the same reason noted in Section 4: these variables are not normally distributed.

6. Key Business Insights

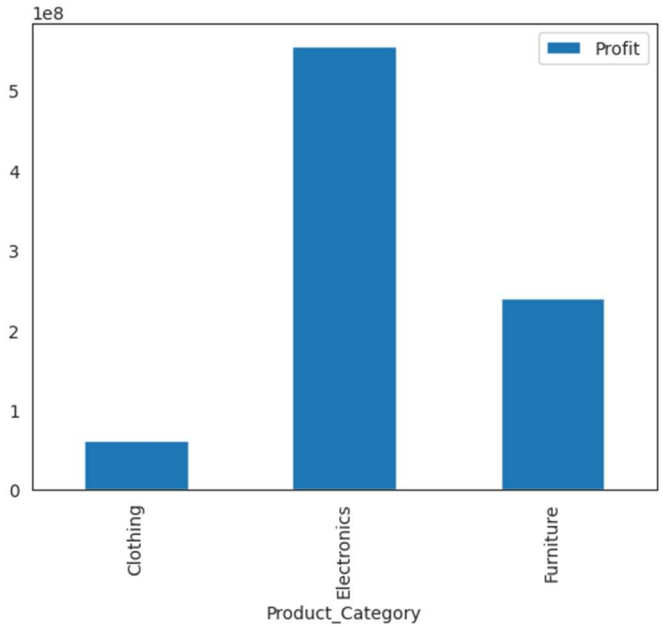
6.1 Most Profitable Region---



Total profit by region (ascending)

Finding: North is the most profitable region (~₹217M), followed by East, West, and South (~₹211M) — a narrow spread of roughly 3% between the strongest and weakest region.

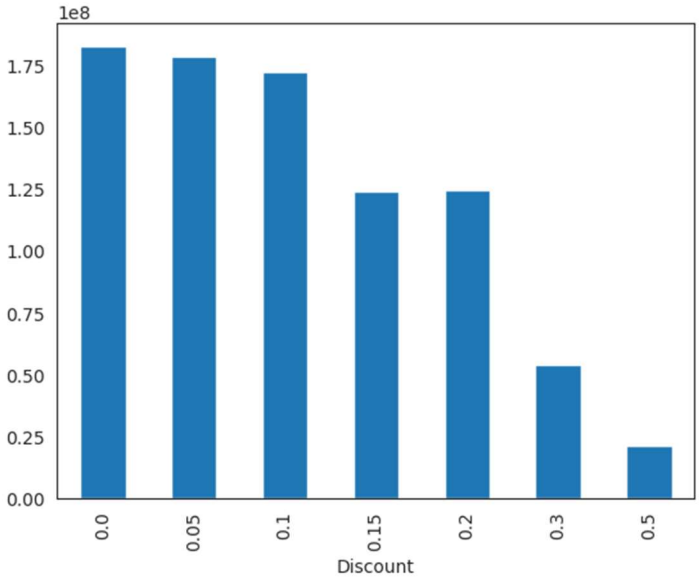
6.2 Product Category to Promote---



Total profit by product category

Finding: Electronics generates by far the highest total profit (~₹555M), well ahead of Furniture (~₹240M) and Clothing (~₹65M), making it the clear priority for continued investment and promotion.

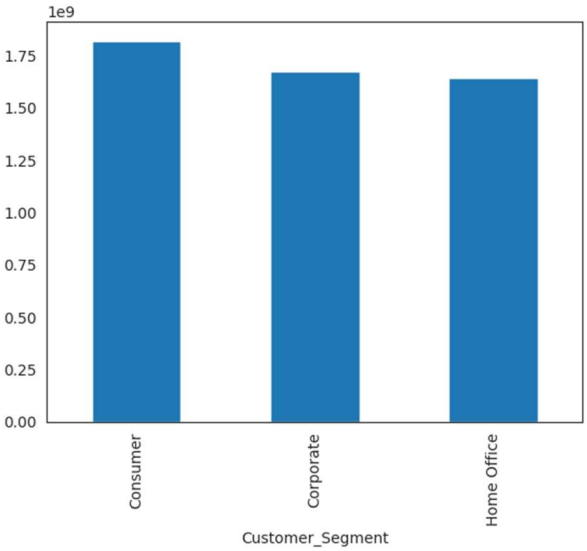
6.3 Discount Strategy vs. Profitability---



Total profit by discount level

Finding: The current discount strategy is not substantially hurting overall profitability, because deep discounts (30–50%) are used on a small share of orders. Profit still declines steadily as discount level rises, so the strategy works only because heavy discounting stays selective — not because discounting is profit-neutral.

6.4 Customer Segment Generating Most Revenue---



Total sales by customer segment

Finding: Consumer is the leading revenue segment (~₹1.82B), with Corporate and Home Office close behind — see Section 5.5 for the full AOV comparison.

6.5 Shipping Cost's Impact on Margins---



Correlation between shipping cost and profit

Finding: Shipping cost has virtually no impact on profit margins (r ≈ 0.003). Profitability is driven by sales value, product mix, and discount level rather than logistics expense.

7. Business Recommendations

This Exploratory Data Analysis (EDA) project successfully examined the e-commerce sales dataset to understand its structure, improve data quality, and identify meaningful business insights. A systematic data preprocessing workflow was carried out by removing duplicate records, handling missing values, correcting inconsistent entries, standardizing categorical variables, engineering new features, and treating outliers. These preprocessing steps enhanced the accuracy, consistency, and reliability of the dataset, making it suitable for detailed analysis.

The exploratory analysis revealed several important business patterns. Sales exhibited clear seasonal trends, with **December** emerging as the strongest-performing month, while **February** consistently recorded the lowest sales. Quarterly analysis showed that **Q2 generated the highest sales**, whereas **Q1 experienced comparatively weaker performance**, highlighting opportunities for targeted marketing and operational improvements. Regional, customer segment, product category, and profitability analyses further demonstrated that business performance varies across different dimensions, emphasizing the need for data-driven decision-making.

The study also highlighted the importance of maintaining high-quality business data. Proper handling of duplicates, missing values, and outliers ensured that statistical summaries and visualizations accurately reflected business performance. Feature engineering enabled meaningful time-based analyses, supporting the identification of seasonal demand patterns and operational trends.

The insights generated from this EDA provide practical recommendations for improving inventory planning, optimizing promotional strategies, enhancing customer engagement, and supporting sales forecasting. By leveraging these findings, organizations can make more informed strategic decisions, improve operational efficiency, and strengthen overall business performance.

In conclusion, this project demonstrates how Exploratory Data Analysis serves as a critical first step in the data analytics lifecycle. Beyond summarizing historical business performance, the cleaned dataset and derived insights establish a strong foundation for advanced analytics, including predictive sales forecasting, customer segmentation, demand prediction, and machine learning applications. The outcomes of this study illustrate the value of data-driven decision-making in enhancing business growth and long-term competitiveness.

8. Conclusion

After cleaning 100,500 raw records down to 100,000 reliable transactions, this analysis shows a business with broadly healthy, evenly distributed demand: sales are spread across regions, states, and customer segments rather than concentrated in a narrow base. Profit is driven overwhelmingly by sales value and product category — Electronics in particular — while shipping cost and order timing have little bearing on margins. Discounting reduces profit per order but is currently applied selectively enough to limit its overall damage. The clearest opportunities are to double down on Electronics, tighten discount discipline at the high end, close the modest regional profit gap led by the South, and lean into the recurring December-peak/February-trough seasonal cycle for planning.